

## Problem-Solution Fit

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|----------------------|--------------------------------------------------------|
| <b>Date</b>          | 27 june 2026                                           |
| <b>Team ID</b>       |                                                        |
| <b>Project Name</b>  | HDI Prediction System using Machine Learning and Flask |
| <b>Maximum Marks</b> | 5 Marks                                                |

### Problem-Solution Fit Canvas:

The Problem–Solution Fit Canvas helps identify the key challenges faced by users and evaluates how the proposed HDI Prediction System addresses those challenges. It analyzes customer needs, existing limitations, root causes, and available solutions to ensure the system provides accurate predictions, meaningful insights, and effective support for decision-making..

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| <b>1. CUSTOMER SEGMENT(S) [CS]</b> <ul style="list-style-type: none"> <li>Government organizations</li> <li>Researchers and policy makers</li> <li>NGOs and development agencies</li> </ul>                                                                                                                                                         | <b>6. CUSTOMER LIMITATIONS<br/>EG. BUDGET, DEVICES [CL]</b> <ul style="list-style-type: none"> <li>Limited technical knowledge</li> <li>Incomplete or inconsistent datasets</li> <li>Budget constraints for advanced tools</li> <li>Limited access to reliable analysis systems</li> </ul>                                                               | <b>5. AVAILABLE SOLUTIONS<br/>PROS &amp; CONS [AS]</b> <p><b>Pros:</b></p> <ul style="list-style-type: none"> <li>Statistical analysis tools exist</li> <li>Public datasets are available</li> <li>Basic visualization options</li> </ul> <p><b>Cons:</b></p> <ul style="list-style-type: none"> <li>Low prediction accuracy</li> <li>Manual processing is time-consuming</li> <li>Limited real-time insights and visualization</li> </ul> |
| <b>2. PROBLEMS / PAINS<br/>+ ITS FREQUENCY [PR]</b> <ul style="list-style-type: none"> <li>Difficult to predict HDI accurately</li> <li>Manual analysis takes a lot of time</li> <li>Lack of data-driven decision support</li> <li>Inconsistent and complex data</li> <li>Limited visualization of results</li> </ul> <p><b>Frequency:</b> High</p> | <b>9. PROBLEM ROOT /<br/>CAUSE [RC]</b> <ul style="list-style-type: none"> <li>HDI depends on multiple complex socio-economic factors</li> <li>Lack of intelligent prediction systems</li> <li>Data scattered across different sources</li> <li>Manual computation and analysis</li> <li>No unified platform for prediction and visualization</li> </ul> | <b>7. BEHAVIOR<br/>+ ITS INTENSITY [BE]</b> <ul style="list-style-type: none"> <li>Collects data from various sources</li> <li>Analyzes data using spreadsheets or reports</li> <li>Seeks accurate predictions for planning</li> <li>High dependency on manual methods</li> </ul> <p><b>Intensity:</b> High</p>                                                                                                                            |
| <b>3. TRIGGERS TO ACT [TR]</b> <ul style="list-style-type: none"> <li>Need for faster and accurate predictions</li> <li>Better policy making and planning</li> <li>Increasing availability of data</li> <li>Demand for data-driven insights</li> </ul>                                                                                              | <b>10. YOUR SOLUTION [SL]</b> <ul style="list-style-type: none"> <li>AI-based system to predict HDI accurately</li> <li>Uses machine learning models on reliable datasets</li> <li>Provides instant predictions and insights</li> <li>Interactive dashboard with visual reports</li> <li>Helps in better decision making and planning</li> </ul>         | <b>8. CHANNELS OF BEHAVIOR [CH]</b> <p><b>ONLINE:</b></p> <ul style="list-style-type: none"> <li>Government portals</li> <li>Research websites</li> <li>Web application / dashboards</li> <li>Social media &amp; community forums</li> </ul> <p><b>OFFLINE:</b></p> <ul style="list-style-type: none"> <li>Government offices</li> <li>Research reports and publications</li> <li>Surveys and field data collection</li> </ul>             |
| <b>4. EMOTIONS BEFORE / AFTER [EM]</b> <p><b>Before:</b> Confused, Uncertain, Frustrated, Overwhelmed</p> <p><b>After:</b> Confident, Satisfied, Informed, Empowered</p>                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                            |

